Day 17 — Timer Event

1. What Is the Timer Event?

> **Timer event:** in the Main Integration Process, the normal way to start a process is the **Start Message** event (a sender triggers it). The **Timer** (from the Palette) is the alternative: it lets the flow **start by itself, without anyone's intervention**, on a schedule.

Key idea: while using the Timer event, **we don't start the message — instead, we schedule the flow: when to start.**

Where Can the Timer Be Used?

Question from Class	Answer
Can we use the plain **Start Event** in the Main Integration Process?	**No ✗** — the Start Event is for Local Integration Processes (Day 16)
① Can we use the Timer in a **Local Integration Process**?	**No**
② Can we use the Timer in an **Exception Subprocess**?	**No**

So the Timer works **only as the start of a Main Integration Process** — this matches the matrix in your Day 16 notes.

2. Timer Start — Configuration

The Timer Start event has two tabs: **General** and **Scheduler**. Inside **Scheduler** there are two modes — **Basic** and **Advanced**.

Basic Mode

Setting	Options (from class)	Meaning
Enter As (frequency)	**Simple Schedule** / **Multiple Schedule**	Simple = one schedule rule; Multiple = several specific dates/times in one timer
Repeat	Seconds, Minutes, Hourly, Daily, Weekly ...	How often the flow should run again
Schedule	**On Deployment** (and a specific **On Day** date/time)	When the first run happens
Time Zone	Choose from the list	The time zone the schedule is evaluated in

"If We Select None" — On Deployment

> **If we select *None* (Repeat = None) with *Schedule: On Deployment*, then when we deploy the flow, message processing starts automatically — once.**

> **Verified against SAP documentation:** "On Deployment" with no repeat runs the integration flow **once, at the moment it is deployed**. "On Day" with no repeat runs it **once at the date and time you specify**. This is the easiest way to test a timer iFlow: set it to run on deployment, deploy, and check Monitor.

Advanced Mode

The **Advanced** option (newer Timer version) is for **complex, granular schedules** — for example, "on the 5th and 20th of every month at 8 AM in a chosen time zone". If you select Advanced, a recurring every-5-minutes schedule is pre-filled by default.

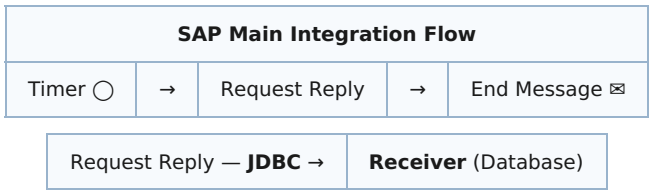
> **Note:** SAP's own blog (when Advanced was introduced) listed typing a raw **cron expression** directly as a future roadmap item — so check your tenant's Timer version to see whether it is available now.

3. Real-Time Use Cases

"In some situations we want to use the Timer event in real time."

Use Case ① — JDBC Adapter (Reading from a Database)

Consider the **JDBC adapter — it will connect to the database.** A database cannot push messages to CPI by itself, so CPI must go and read it on a schedule.



Reading the flow: the Timer starts the flow on schedule → a **Request-Reply** step calls the database through the **JDBC** adapter and brings the data back → the flow ends with **End Message**.

> **Tie-ins:** the database connection details are stored as **JDBC Material** in the Manage Security tab (Day 6). And combining this with Day 13's **variables** gives you a proper **delta load** — store the last-run timestamp, and on every timer run read only the rows changed since then.

Use Case ② — Pulling Data from an External Source Yourself

> **If we want to hit some external source and get the data by ourselves, then we can use the Timer event.**

Push (Start Message)	Pull (Timer)
The **sender** decides when to send; CPI waits at an endpoint	**CPI** decides when to go and fetch the data from the source
Needs a Sender channel (e.g., HTTPS)	No Sender channel — the schedule is the trigger

4. Why Is There No Endpoint After Deploying a Timer iFlow?

Observation: there won't be any endpoint after triggering the iFlow in **Monitor → Integration Content**.

> **Why? Answer:** because we have **not configured any Sender connection adapter**, CPI **won't generate any endpoint** in the Monitor tab.

This also means **you cannot trigger a timer iFlow from Postman** — there's nothing to call. You trigger it with the schedule (or by deploying it with "On Deployment"), and then check the result under **Monitor → Message Processing**.

5. Practical Tips (Added)

- **The Timer starts with an empty message** — there is no incoming payload. Build your body with a Content Modifier, or let the Request-Reply bring the data in.
- **Set the Time Zone deliberately** — a schedule "daily at 8 AM" means 8 AM *in the zone you choose*.
- **Be careful with very frequent schedules** (every few seconds) — every run creates a message log, which floods Monitor and uses tenant capacity. Use sensible intervals while practicing in your trial tenant.
- **To stop a running timer iFlow**, undeploy it (Monitor → Manage Integration Content) — otherwise it keeps running on schedule.
- **"On Deployment" re-runs on every redeploy** — redeploying after a small change will trigger one more run.

6. Quick Interview Answers

Question	Short Answer
When do you use a Timer start event?	When the iFlow must run on a schedule, or must **pull** data (e.g., from a database or API) instead of waiting for a sender to push
Can a Timer be used in a Local Integration Process or Exception Subprocess?	**No** — only to start a Main Integration Process

Why does a timer iFlow show no endpoint in Monitor?	It has **no Sender channel**, so no endpoint is generated
How do you run a timer iFlow exactly once?	Repeat = **None** with Schedule = **On Deployment** (or **On Day** for a specific date/time)